FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Student Assessment Document

FINAL EXPLANATION: GENERAL SCIENCE – GRADE 10 | SUB-STRAND 2.3: CHEMICAL BONDING	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS

WHAT IS THIS DOCUMENT?

This is your Final Explanation guide. By the end of this unit, you should be able to write a complete, evidence-based answer to the driving question using everything you have learned across all 8 lessons. Read this document carefully before you begin writing your response.

YOUR DRIVING QUESTION:

"Why do the aluminium sufuria, the graphite pencil and the table salt on our kitchen table look similar as solids but behave so differently — and what is happening between the atoms inside each one that controls every property we can measure?"

HOW TO USE THIS DOCUMENT:

- Read each section prompt carefully before you write.
- Use the exemplar paragraphs as a model — notice how they link particle-level evidence to observable properties.
- Do NOT copy the exemplars. Use them to guide the structure and depth of your own answer.
- Connect every claim you make back to the anchoring phenomenon (the three items on the demonstration desk).
- Use scientific vocabulary accurately: bond type, lattice, delocalised electrons, electrostatic attraction, layers, ions, melting point, conductivity, malleability.
- Aim for a complete response that addresses all five sections.

Section 1 — Identify the Bond Type in Each Substance	
For each of the three kitchen substances — table salt (NaCl), aluminium (Al), and graphite (C) — identify the type of chemical bonding present. Explain what particles are involved in each bond and how the bond forms. Why is it important to identify	The three substances on the demonstration desk look similar as solids, but they are held together by completely different types of chemical bonds — and that difference is the key to everything else we observe. Table salt (NaCl), harvested from the Kenyan coast, contains an ionic bond. This forms when sodium transfers one electron to chlorine, producing a positive Na^{+} ion and a negative Cl^{-} ion. These oppositely charged ions attract each other strongly through electrostatic forces. Aluminium metal contains a metallic bond: the aluminium atoms release their outer electrons into a shared 'sea' of delocalised electrons that flows freely between the positive Al^{3+} ions. Graphite is a form of carbon held together by covalent bonds. Each carbon atom forms three strong covalent bonds with its neighbours in a flat, hexagonal layer. The fourth outer electron from each carbon atom is delocalised across the entire layer. Identifying the bond type matters because the type and strength of the bond directly controls every measurable

the bond type before explaining any physical property?	property — melting point, conductivity, and mechanical behaviour — as we will see in the sections below.

Section 2 — Structure at the Particle Level	
Describe the three-dimensional arrangement of particles (the structure) inside each substance. How does this internal structure differ between table salt, aluminium, and graphite? Use the terms 'lattice', 'layers', and 'sea of electrons' where appropriate, and connect each structure back to what you could see or do to the sample on the demonstration desk.	The internal structure of each substance explains the shapes and textures we observed on the demonstration desk. Table salt forms a giant ionic lattice: millions of Na^+ and Cl^- ions are arranged in an alternating, three-dimensional grid, held in fixed positions by strong electrostatic attractions in all directions. This is why the salt crystal is rigid and cannot be bent — the ions are locked in place. Aluminium also forms a giant metallic lattice: rows of Al^{3+} ions are packed closely together and surrounded by the free-moving sea of delocalised electrons. Because the electron sea holds the ions together without locking them into one fixed arrangement, the layers of ions can slide over each other when force is applied — which is why the sufuria wire bent without snapping. Graphite has a layered covalent structure: flat sheets of carbon atoms, each bonded in a hexagonal network, are stacked on top of each other. The bonds within each layer are strong, but the forces between the layers are very weak, allowing layers to slide over one another easily. This is exactly why the graphite rod left a grey mark on the paper — thin layers flaked off and stayed behind.

Section 3 — Explaining Conductivity (The Circuit Experiment)	
When the teacher connected each substance to a simple circuit with a bulb, the results were surprising. Explain, using particle-level reasoning, why aluminium conducts electricity as a solid, why dry table salt does NOT conduct as a solid but DOES conduct when dissolved in water, and why graphite — a non-metal — also conducts electricity. What does this tell us about what is needed for a substance to conduct electricity?	The circuit experiment was the most surprising part of the demonstration, and bond type explains every result perfectly. For a substance to conduct electricity, it must contain charged particles that are free to move and carry charge through the material. Aluminium conducts electricity as a solid because its delocalised electrons are always free to drift through the metallic lattice when a voltage is applied. This is why the bulb lit up immediately when the sufuria wire was placed in the circuit. Dry table salt does NOT conduct electricity because, in the solid ionic lattice, the Na^+ and Cl^- ions are locked in fixed positions by the strong electrostatic forces all around them — they cannot move. However, when salt is dissolved in water, the lattice breaks apart and the ions become free to move through the solution. These mobile ions carry charge and the bulb lights up. This shows that ions can conduct electricity, but only when they are free to move. Graphite conducts electricity even though carbon is a non-metal because each carbon atom contributes one delocalised electron to the layer. These electrons are free to move across the entire sheet of carbon atoms — just like the electrons in a metal. This is why graphite is used as an electrode in batteries and electrolysis cells. The rule we can write is: electrical conductivity requires freely moving charged particles — either delocalised electrons (as in metals and graphite) or free ions (as in ionic solutions).

Section 4 — Explaining Melting Point and Mechanical Behaviour	
The teacher showed that table salt melts at above 800 °C, aluminium melts at 660 °C, and graphite layers flake off easily	To melt a solid, you must supply enough energy to overcome the forces holding the particles together. The stronger the forces, the higher the melting point. Table salt has the highest melting point of the three (above 800 °C) because the electrostatic attractions between the Na^+ and Cl^- ions act in all three directions through the lattice and are extremely strong — a huge amount of thermal energy is needed to pull every ion away from its neighbours. A home jiko burning charcoal in Nairobi cannot produce enough heat

even at room temperature. Also, the aluminium wire bent without breaking, but the graphite rod snapped when force was applied. Use your knowledge of bond type and structure to explain ALL of these observations. What must happen at the particle level for a solid to melt or to be deformed?	to melt salt, which is why it stays solid during cooking. Aluminium melts at a lower temperature (660 °C) because the metallic bonds, though still strong, involve the electrostatic attraction between the electron sea and the Al^{3+} ions in only one type of interaction, and these can be overcome with less energy than the full ionic lattice. For mechanical behaviour, the aluminium wire can bend because the layers of Al^{3+} ions can slide past each other while still being held by the same electron sea — the bond type does not break. This property is called malleability. Table salt, however, will shatter rather than bend: if you push the layers of ions sideways, ions of the same charge are forced next to each other, the repulsion is enormous, and the crystal breaks apart. Graphite flakes easily because the covalent bonds within each carbon layer are very strong, but the forces between the layers are weak. A small force is enough to slide one layer off another — which is why the graphite rod left a mark on paper and why the rod was brittle and snapped when bent.

Section 5 — Answering the Driving Question (Your Complete Explanation)	
Now bring everything together. Write a full, connected answer to the driving question. Your answer must: (1) state the three bond types and the three structures; (2) use particle-level reasoning to explain conductivity, melting point, and mechanical behaviour for ALL three substances; (3) explain what was puzzling about the phenomenon at the start and how your understanding of chemical bonding now resolves that puzzle; and (4) state one real-world application or implication of what you have learned — you may use Kenyan examples.	At the start of this unit, we faced a genuine puzzle: three everyday kitchen items — table salt from the Kenyan coast, an aluminium sufuria wire, and a graphite rod from a used battery — all appeared as dull or shiny solids, yet they behaved in completely opposite ways when we tested them for conductivity, melting point, and mechanical strength. The answer to every part of this puzzle lies in the type of chemical bond holding the atoms or ions together, and the three-dimensional structure that bond creates. Table salt (NaCl) contains an ionic bond: Na^{+} and Cl^{-} ions are held in a giant three-dimensional lattice by strong electrostatic attractions in all directions. Because the ions are locked in fixed positions, dry salt cannot conduct electricity. Only when dissolved in water do the ions become free to move and carry charge. The strong, all-directional forces also require over 800 °C to overcome, giving salt its very high melting point, and the rigid lattice makes it hard but brittle. Aluminium contains a metallic bond: a sea of delocalised electrons surrounds a lattice of Al^{3+} ions. The free electrons allow aluminium to conduct electricity immediately, even as a solid. The metallic bond is strong enough to give aluminium a melting point of 660 °C, but because the electron sea maintains the bond even as ion layers slide past each other, aluminium is malleable — it bends without breaking. This is exactly why aluminium is used to make sufuria handles and cooking pots across Kenya. Graphite is a covalent structure: each carbon atom forms three strong covalent bonds within a flat hexagonal layer, and one delocalised electron per atom is free to move across the layer. This makes graphite a conductor of electricity — surprising because carbon is a non-metal. Between the layers, the forces are very weak, so layers slide off easily, leaving marks on paper and making graphite useful as the 'lead' in pencils. The layers also make graphite brittle — it snaps rather than bends. The driving question asked what is happening between the atoms inside each solid that controls every property we can measure. The answer is: bond type determines the arrangement of particles, and the arrangement of particles determines every property. There are no exceptions among the three substances. One important real-world implication is in recycling: aluminium from old sufuria and cooking pots can be melted down and reformed at just 660 °C, making it energy-efficient to recycle in Kenyan industries. Salt, by contrast, would require far more energy to melt, and graphite is used in the electrodes of the very electrolysis cells that are used to purify metals in chemical industries. Understanding chemical bonding is not an abstract exercise — it explains the materials that Kenyans use, recycle, and depend on every day.

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Bond Type Identification	Correctly identifies ionic bonding in NaCl, metallic bonding in Al, and covalent (+ delocalised electrons) in graphite. Accurately describes the particles involved and how each bond forms, with no errors.	Correctly identifies all three bond types and mostly describes the particles involved, with one minor error or omission in the description of bond formation.	Identifies one or two bond types correctly but confuses the particles involved or omits a key bond type entirely.
Particle-Level Structure	Accurately describes the giant ionic lattice (NaCl), metallic lattice with electron sea (Al), and layered covalent structure (graphite). Correctly uses 'lattice', 'layers', and 'sea of electrons' and connects each structure directly to observable features of the substances on the demonstration desk.	Describes the structure of all three substances with mostly accurate detail. Uses at least two of the required terms correctly and connects at least two structures to the observed phenomenon.	Describes the structure of only one or two substances, or uses structural terms incorrectly. Makes limited connection to the observed phenomenon.
Explaining Conductivity Using Evidence	Provides a complete, accurate particle-level explanation for why Al conducts as a solid (free electrons), why dry NaCl does not conduct but conducts in solution (ions locked vs. free), and why graphite conducts (delocalised electrons across layers). Clearly states the general rule about freely moving charged particles.	Explains conductivity for all three substances with mostly correct particle-level reasoning. May omit the general rule or provide a partially incomplete explanation for one substance.	Explains conductivity for only one or two substances, or relies on surface-level observations (e.g., 'it is a metal') without particle-level reasoning.
Explaining Melting Point and Mechanical Behaviour	Accurately links the high melting point of NaCl to strong all-directional electrostatic forces, Al's lower melting point to the metallic bond, and graphite's flaking to weak interlayer forces. Correctly explains malleability of Al (sliding ion layers in electron sea) and brittleness of NaCl and graphite using particle-level arguments.	Correctly explains melting point trends and mechanical behaviour for at least two of the three substances using particle-level reasoning. One explanation may lack full detail.	Describes the observed behaviours (e.g., 'salt has a high melting point') without connecting them to bond type or particle-level structure, or provides only one correct particle-level explanation.
Synthesis and Real-World Application	Writes a coherent, complete answer to the driving question that integrates all three bond types, all three structures, and all tested properties. Clearly articulates how chemical bonding resolves the original puzzle. Provides at least one accurate, relevant real-world application using a Kenyan context (e.g., aluminium recycling, graphite electrodes, salt in industry or cooking).	Integrates most of the unit's concepts into a connected answer. Addresses the original puzzle and includes a real-world application, though the connection to a Kenyan context may be general or partially developed.	Provides a partial synthesis that addresses some but not all of the properties or substances. The answer to the driving question is incomplete, and the real-world application is absent or not connected to chemical bonding concepts.